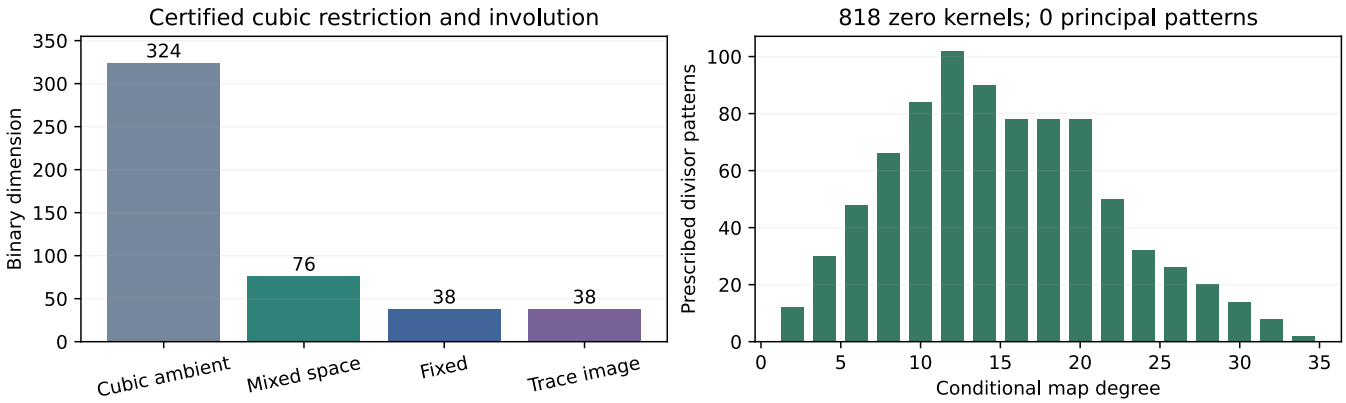
R61: complete cubic space and 818 support certificates

The actual cubic ambient space has dimension 324. All 248 local constraints leave the complete mixed space of dimension 76. Its fixed space and trace image both have dimension 38. All 38 actual fixed functions,38 trace preimages and a trace-one function satisfy whole-source identities.



Recorded dimensions: cubic61-02 and fix61-02. The histogram aggregates all 818 rows of cohort61-01. 818 patterns have zero kernel; 0 have one-dimensional principal-function spaces. The fresh checker reconstructs every matrix from ambient profiles before fixed projection.

Complete cubic construction and source binding

The 324 descended cubic products are checked on all 20 normalized charts: 6480 pole bounds and 1360 old-function coefficient conjugations. All 385 F4 regular jets give770 binary rows and an explicit left inverse. The ambient degree 384 and genus 61 Riemann-Roch bridge imply completeness. All 248 binary conditions have rank248; full C K=0, R K=I76 and K R+H C=I324 certify the degree 136 mixed space, with caps 34,12,20,2 at four degree 2 infinity points. All 324 cubic and 76 mixed functions are bound in the actual degree 32 function field, including all 76 involutions.

With N=sigma+I, the actual fixed/trace complex has an explicit Jordan basis, both inverse products and a contraction Q satisfying NQ+QN=I76, Q squared=0. All fixed local and regular profile imaginary components vanish. Both trace parities remain available before the principal-x gate.

Complete prescribed support cohort

Every triple in the intervals [−6,6],[−10,10],[−1,1] fixes the first signed ramification by balancing; only the all-zero triple is erased. All 818 cases impose 136 binary coefficient rows on the complete 38-space. Each zero kernel has a saved H C=I38. Any one-dimensional kernel has C K=0, R K=1 and K R+H C=I38, with its generator retained. The global regular positive control has rank 37 and kernel 1 containing actual 1. Map degrees are at most 34, including degrees divisible by 11.

The checker reconstructs constraints in the 324-dimensional ambient profiles, then applies the mixed and fixed inclusions. It also checks the ambient injection, full mixed/fixed splitting, contraction, odd particular, regular control and a changed-row negative control. Proper-source, normalization, divisor and Riemann-Roch identifications remain explicit geometric premises; the certified cohort is the stated pole support.

Execution, formal implications and the next geometry

Attempt	Outcome	Native wall
check61-01	rejected	482 s
check61-02	rejected	1 s
check61-03	rejected	491 s
cohort61-01	accepted	328 s
cubic61-01	rejected	194 s
cubic61-02	accepted	829 s
field61-01	accepted	5 s
fix61-01	interrupted	not recorded
fix61-02	accepted	5 s
rustcheck61-01	accepted	same UTC second
scaling-lean61-01	accepted	not recorded
trace61-01	accepted	271 s

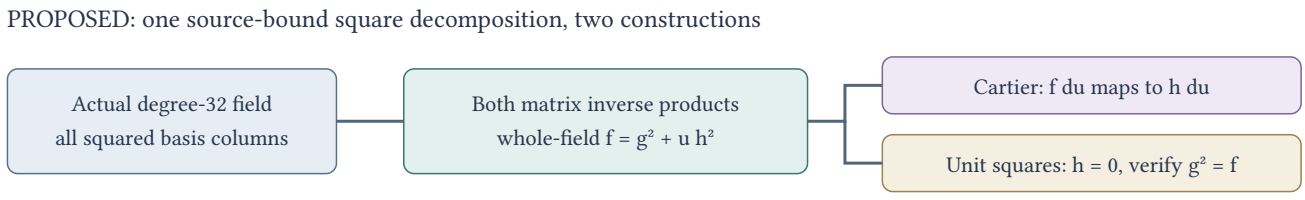
Native wall is derived from UTC timestamps with one-second resolution. A same-second result does not measure zero work; compiler wall was not recorded separately. The listed interval covers only that invocation; it is not a performance comparison. Each raw directory preserves its exact source revision, copied source/protocol, native binary hash and output. Seed61; one native job/thread;1200seconds and4GiB virtual per invocation. Lean4.34.1/mathlib d13f23b723b8a846827a245b89c10fc7d3f11612, -j1 -M4096, separate1200second compile/audit limits and no virtual cap.

The first cubic attempt reversed the retained section/quadratic pair indices, produced native errors and was stopped. Its exit0 does not make its output acceptable. The repair was frozen before fresh02. The first fixed-space launch was interrupted at the checkpoint gate during migration, before native start; all saved inputs are retained. Fresh02 used unchanged mathematical source. The first checker was manually stopped after an avoidable per-cell interpreter loop was identified; it has no accepted result and is not labelled a timeout. Checker02 then failed on unsupported module(int) zero initialization; its late PASS and numbers are rejected. Fresh03 removes that reset but was resource-stopped during the generic polynomial-matrix products. The accepted complete Rust replay instead uses packed binary row-dot-column parity for all 818 pre-projection products, all source/involution/Jordan/trace/control products and changed-inverse plus changed-row controls. All resource stops are distinct from timeouts. No accepted producer was repeated for publication. Historical auxiliary protocol metadata defects and prospective repairs are disclosed in PROTOCOL-METADATA61.md; raw versions remain immutable.

Six audited Lean implications

The six Scaling61 declarations prove arbitrary-cap balanced first and conditional degree bounds, a symbolic product-minus-one cohort count, the 818 corollary, inverse squaring from x to power32=x, and obstruction to surjectivity of C-I from a nonzero invariant functional. The source geometry and Cartier interpretation are external premises, separate from these generic kernel-checked implications. Compile/audit receipts identify all six declarations and their axiom status. Both exits are zero, and the audit reports six theorems, zero failing and zero trusted.

Trace dual and new construction paths



Design: SQUARE-DECOMPOSITION61.md. Holomorphic regularity, full divisors and target action remain separate checks.

All 528 normal-product columns and 1024 ordered trace products give a rank 32 trace pairing. Both inverse products and 32 actual source-coordinate dual columns are certified, with a changed-Gram negative control. This is the affine trace-dual lattice; global regular differentials require the complete infinity conditions described in DIFFERENTIAL-DESIGN61.md.

The next differential construction uses a certified normal-basis pole bound to obtain a complete finite coefficient span, imposes every infinity regularity condition, then checks the full 61-dimensional holomorphic space and source-bound Cartier/sigma operators. CARTIER-CONTROL61.md records the Artin-Schreier quotient's nonzero joint fixed-space control and convention checks. CUSPIDAL-UNIT-LEAD61.md gives a second path through exact generated unit lattices and characteristic2 saturation. Neither proposed path is reported as an executed differential or saturation result here. SQUARE-DECOMPOSITION61.md proposes one certified 32x32 square-basis inverse for both paths, with exact rational coefficient splitting and source checks.